

Privacy-First Dashcam Vehicle Recognition

A self-built computer-vision system that reads traffic — make, model, generation, plates, and speed limits — fully offline on a MacBook

Executive Summary

I built, from scratch, a privacy-first dashcam system that understands the street: it detects and tracks every vehicle in real time, identifies make + model + generation offline on Apple Silicon, reads license plates (including DK/DE electric and historic-plate detection), recognizes speed-limit signs, and blurs every plate for GDPR — all without any cloud API at inference. The system also auto-collects its own training data from every drive and was used to train a 646-class European vehicle classifier through seven verified training iterations.

Why it matters

- Dashcams record, but don't understand. This one reads traffic — and does it privacy-first (plates auto-blurred on stream and recordings).
- The whole pipeline — data collection, labeling, training, evaluation, deployment — runs on a single MacBook plus rented GPU hours.
- Built-in honest evaluation methodology: a permanent frozen holdout that no training build may ever touch.

System Capabilities

Capability	Implementation	Detail
Vehicle detection + tracking	YOLOv8 + ByteTrack	~25 ms/frame on Apple Silicon (MPS)
Make/model/generation (offline)	ConvNeXt-Tiny classifier	646 European classes, from scratch
License plates	EasyOCR + heuristics	DK/DE plates, electric (E) / historic (H) tags
Speed-limit signs	GTSRB classifier	EU signs, two-read confirmation
Privacy	GDPR auto-blur	plates blurred on stream + recordings, on by default
Live dashboard	FastAPI + WebSockets	localhost:8500, near-fullscreen driving UI
Dataset growth	Auto crop harvesting	every drive labels new crops for retraining

Model & Results

The deployed classifier is a ConvNeXt-Tiny (28M parameters) trained from scratch on 646 European vehicle classes. It runs fully offline via MPS inference with test-time augmentation.

Honest evaluation

Every number below was measured on an intersection of permanent holdout sets: 597 real dashcam crops that **no model ever trained on**. This methodology was built deliberately — naive holdout comparisons were found to be contaminated because models had seen parts of each other's eval sets.

Model	Top-1	Top-5	Verdict
v19 (deployed king)	57.2%	79.7%	Best honest model
v20 (+5k crops)	56.5%	79.4%	Tied — data plateau
v21 (frozen-holdout)	54.7%	77.9%	Honest baseline
v22 (+1k recovered)	53.7%	79.9%	Recovery = noise

Engineering journey — what actually moved the needle

- **30x oversampling was a trap:** the model memorized duplicate crops (97% internal validation → 44% on real holdout). 5x + from-scratch fixed it.
- **Data diversity beats architecture:** tiny/base/large and 224px/336px were all tested; tiny @ 224px won every time.
- **Evaluation methodology matters as much as the model:** a track-level leak was found and fixed; a permanent frozen holdout now guarantees honest comparisons forever.
- **More of the same data has a hard plateau** — proven with three consecutive experiments. The next real gain is new-domain footage (own recordings), not more similar YouTube data.

Training scale

Metric	Value
Dataset (raw labeled crops)	~21,300 crops
Training set (5x oversample)	~80,000 images
Classes	646 deployed / 822 total
Training rounds (verified)	7 (v15→v22)
Rented GPU	RTX 4090 / 5090 (RunPod)
Local hardware	MacBook M5, 24 GB unified

Pipeline & Tooling

- `extract_crops` — mines unique crops per car per distance bucket from video.
- `label_crops` — batch labeling via DeepSeek vision (paid, no quota).
- `build_merged` — dataset builder: canonicalization, group-aware holdout, oversampling, leak-free frozen eval set.
- `train_classifier` — from-scratch training (MPS local or CUDA pod).
- `compare_ckpts` / `eval_all` — honest holdout + intersection evaluation.
- `main.py` + frontend — live FastAPI dashboard for driving.
- Grok/Gemini — second-opinion reject recovery for hard crops.

Privacy & Ethics

- Plates are blurred on stream and recordings (GDPR), on by default.
- Plates used for in-memory vehicle re-ID only — never streamed or stored.
- Training data derived from publicly available dashcam footage; the public dataset card documents provenance under a research license.

Tech Stack

Python 3.14 · PyTorch (MPS + CUDA) · ConvNeXt · YOLOv8 · ByteTrack · OpenCV · FastAPI + WebSockets · EasyOCR · GTSRB · yt-dlp · RunPod · DeepSeek/Gemini/Grok vision APIs

Built from scratch: video mining → labeling → dataset engineering → training → honest evaluation → deployment → live dashboard. All on a MacBook.

Report generated 2026-08-31 · Full code: [github](#) (see repo) · Model + dataset: [Hugging Face](#) (see cards)